Sustainable Aviation Fuel Supply Roadmap for Korea

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Objective and contribution

SAF supply planning requires matching domestic feedstocks and conversion routes with future fuel demand and the carbon and hydrogen needed for synthesis. I led an eight-member course team in developing a Korean supply roadmap, using European blending targets as a scenario benchmark.

Supply and demand analysis

We compared hydroprocessed esters and fatty acids (HEFA), alcohol-to-jet (ATJ), gasification/Fischer–Tropsch (FT), and power-to-liquid (PtL) pathways. To estimate supply potential from domestic resource inventories, we used feedstock heating values and conversion efficiencies for bio-SAF and carbon-content calculations for synthetic SAF. We projected jet-fuel demand through 2050 using linear regression on 2009–2021 consumption data and applied the benchmark blending targets.

Resource or input	Route	Estimated annual potential
Rice straw	ATJ	506.8 million L (bio-SAF)
Used cooking oil	HEFA	227.8 million L (bio-SAF)
Organic wastes and sewage sludge	PtL	268.0 million L (synthetic SAF)
2050 synthetic-SAF balance (million L/year) Supply: 268.0 | Requirement: 522.4 | Shortfall: 254.3		

Table 1. Annual SAF potential and 2050 synthetic-SAF balance. Values are rounded independently.

Under the report's assumptions, the combined bio-SAF potential of 734.6 million L/year covered the modeled requirement through 2050. The synthetic-SAF potential met the modeled requirement through 2045 but left a projected shortfall of 254.3 million L/year in 2050, prompting consideration of additional carbon resources.

Supply conditions and project outcome

We examined direct air capture (DAC) as an additional CO_2 source and considered hydrogen availability. We compared projected SAF prices with a conventional jet-fuel price benchmark to examine the cost gap over time. The roadmap highlighted feedstock collection systems, additional CO_2 sourcing, and hydrogen supply as conditions for domestic production. The project ranked first among eight teams.